



Control of preferential He permeation with adsorptive molecules at temperatures extensively ranging from ambient to extremely low temperatures

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ABSTRACT

He separation performance of mixtures derived from natural gases, including He, CH₄, and N₂, at significantly low temperatures was investigated using 1,2-bis (triethoxysilyl) ethane (BTESE) bridged-type organosilica membranes with different pore sizes. In this study, the membrane pore size was successfully controlled through varied sol preparation conditions (pH swing, acid/water molar ratio). A BTESE membrane with small pores showed high He permselective property (i.e., He/CH₄: 630, He/N₂: 320 at 200 °C) from 30 to 200 °C owing to enhanced molecular sieving. However, the He permeance of this membrane significantly decreased as permeation temperature decreased and the He selectivity reduced to Knudsen ratio at −115 °C. In contrast to a BTESE membrane with small pores, a BTESE membrane with relatively large pores exhibited low He selectivity from 30 to 200 °C. However, this membrane showed CH₄ or N₂ permselective (CH₄/He: 10, N₂/He: 6 at −115 °C) properties, as well as high CH₄ and N₂ permeance of $1.2 \times 10^{-6} \text{ mol m}^{-2} \text{ s}^{-1} \text{ Pa}^{-1}$ (3600 GPU) and $8.7 \times 10^{-7} \text{ mol m}^{-2} \text{ s}^{-1} \text{ Pa}^{-1}$ (2600 GPU) at −115 °C, respectively. This was caused by surface diffusion of CH₄ and N₂ at significantly low temperatures, which effectively inhibited the permeation of He, which has a smaller molecular size than those of CH₄ and N₂, through adsorption blocking. These results shed light on the separation properties of an adsorptive/non-adsorptive system at extremely low temperatures and recommend that the use of a membrane with large pore sizes is effective, as opposed to small pore sizes, which is the appropriate design at ambient temperature, to exploit the molecular sieving.

1. Introduction

He is the second most abundant constituent in the universe, although its concentration is only 5 ppm on Earth. The presence of He went unnoticed for a long time, until William Ramsay found proof of its existence in ores in 1895. The He that was present on ancient Earth has already diffused into the universe. Therefore, the vast majority of terrestrial He that is found today is generated by the alpha decay of heavy radioactive elements such as uranium within the crust of Earth. Owing to its inherent characteristics, such as its small molecular size, significantly low boiling point of 4 K [1,2], which is ascribed to the induced dipole called the London dispersion force, high-speed diffusion, high thermal conductivity, incombustibility, and chemical stability [3], He has been used in a wide range of industrial fields from detection to magnetic resonance imaging, gas chromatography, and the space

industry. Practically, He is primarily extracted by purifying natural gas because underground deposits contain a higher concentration of He, ranging from 0.05 to 1% as compared to the trace amounts found in the atmosphere. The majority of the natural gas reserves worldwide are held in countries such as Russia, Qatar, and North America, which account for 90% of all global deposits [4,5]. Aside from He, natural gas includes CH₄ (70–90%), C₂ – C₃ hydrocarbons (up to 20%), C₄ hydrocarbons (1–3%), and N₂ (up to 10%). Minor constituents, such as hydrogen sulfide and water, are removed during the pre-purification process to avoid pipe corrosion. He is purified via a cryogenic separation process that is operated at temperatures ranging from −115 to −225 °C [6]. Therefore, the final feed components obtained through this process are CH₄, N₂, and He. Because of the small fraction of He in the feed, which has an extremely low partial pressure, He separation through a stand-alone membrane is not feasible. Consequently, integrated distillation

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and membrane processes [7,8] have been considered after cryogenic purification. In this process, cryogenic distillation initially increases the He concentration from 60 to 90% to secure enough partial pressure for He permeation, and the roughly purified gas stream is purified using He permselective membranes at ambient or elevated temperatures.

With the need for superior He permselective membranes, a wide range of materials including organic polymers, organic–inorganic hybrids, and microporous inorganic membranes have been reported to enhance He permeance and selectivity [9,10]. For instance, organic polymer membranes, which have historically been used for He recovery, have attracted research interest. To enhance He permeability, a perfluorinated polymer is typically coated on a porous support to form a thin layer [11–13]. However, excluding a few polymeric membranes, which exceed the Robeson upper bound of the permeance-selectivity trade-off relation [14], most organic polymer membranes exhibit low He permeance and selectivity. Organic–inorganic hybrid structures, where nanoporous media such as silica and zeolite are incorporated into a polymer structure (mixed matrix membranes (MMMs)), are highly effective in controlling the free volume of a polymer network and forming a diffusion path through nanoporous media [15]. Although He permeability is significantly improved using these hybrid structures, its permeability and selectivity are still lower than that using microporous inorganic membranes [9]. Currently, carbon molecular sieve [16–18], microporous amorphous silica [19–22], and crystalline zeolite [23–26] are representative inorganic materials for membranes that exhibit high He separation performance.

When developing a separation system, such as for He (0.26 nm), N₂ (0.364 nm), and CH₄ (0.38 nm), gases with small kinetic diameter such as He generally become the preferential permeation component and high selectivity of such components can be achieved through the effect of activated diffusion via molecular sieving at ambient or elevated temperatures. However, to reduce the energy consumption of cryogenic distillation itself, in the present study, we propose the introduction of a membrane separation process before cryogenic distillation. In this case, because of the low mole fraction of He, the preferential permeation of adsorptive molecules such as CH₄ and N₂ (which are abundant constituents) can occur at extremely low temperatures, resulting in the concentration of He in the retentate stream. A rubbery polymer membrane can achieve CH₄ permselective property at ambient temperature, demonstrating a solution-diffusion permeation property that favors CH₄ sorption compared to that of He [9]. However, as the permeation temperature decreases below the glass-transition temperature, owing to the operational restriction induced by the transformation from a flexible rubbery structure to a solid glassy structure, the polymer membrane eventually loses the CH₄ permselective property [27].

Rao et al. [28] were the first researchers to demonstrate the surface diffusion of adsorptive molecules through nanoporous carbon membranes at low temperatures. Notably, several fundamental membrane designs for achieving efficient surface diffusion have been proposed. Moreover, experiments on the permeation of hydrogen and C₁–C₄ hydrocarbons have demonstrated two intriguing conclusions. First, in single-gas permeation experiments, a direct correlation between adsorption heat and separation performance was not observed, that is, a higher adsorption heat did not imply higher permeability. Second, increased adsorption heat of the co-existing hydrocarbon enhances the selectivity for hydrogen in a binary separation of hydrogen and a hydrocarbon. However, the lowest temperature in these experimental conditions was –15 °C. Hence, the effect on the separation performance at temperatures lower than –15 °C has not yet been adequately confirmed. Baker et al. [29] theoretically proposed the gas permeation behavior of various gases, including light hydrocarbons, noble gases, some inorganic gases, at significantly lower temperatures up to 190 K through a silicalite-1 membrane and proved that experimental results agreed with the theoretical equations considering surface diffusion. Ye et al. [30] focused on the separation performance of N₂ and He through MFI-type zeolite membranes at temperatures ranging from 85 to 260 K.

Their study demonstrated that an MFI-type zeolite membrane was N₂ permselective (highest performance at 124 K; N₂ permeance: 3.9×10^{-6} mol m⁻² s⁻¹ Pa⁻¹, N₂/He: 75.7) over the entire temperature range in binary permeation. Yu et al. [31] evaluated He selectivity in He/N₂/CH₄ ternary system using MFI membranes under temperatures ranging from 120 to 293 K and concluded the effectiveness of molecular adsorption through membrane pores for achieving high selectivity. Considering the bilateral interaction of gas permeation, the separation mechanisms of various systems at low temperatures were explored. Izumi et al. [32] investigated the separation performance for He/O₂, which is a non-adsorptive/adsorptive system, and O₂/N₂, which is an adsorptive/adsorptive system, at –115 °C using organosilica membranes. They concluded that O₂ permeation is unaffected by the presence of He, whereas O₂ permeation is inhibited by co-existing N₂. Ishizaki et al. [33] demonstrated superior O₂ separation performance for O₂/N₂ using organosilica membranes. The experimental result revealed that finely controlling the membrane pore sizes to enhance the molecular sieving effect is significantly effective in the separation systems, in which only a minor difference exists between the adsorption heats of the components. However, in separation systems where the components have different adsorption heats, such as He, CH₄, and N₂, and the effect of tuning the membrane pore size to enhance molecular sieving at significantly low temperatures has not been investigated.

Therefore, this study attempted to shed light on the control of preferential gas permeation at significantly low temperatures for He separation from CH₄ and N₂ using 1,2-bis (triethoxysilyl) ethane (BTESE) as a bridged-organosilica membrane. Single-gas permeation experiments were performed using membranes with two different pore sizes to determine the dependence on kinetic diameter by using several gases with varying kinetic diameters. Moreover, the temperature dependence of these gases at temperatures ranging from –115 to 200 °C was observed to evaluate the effect of membrane pore size and permeation temperature on the permeation mechanism. To confirm the affinity between the membrane pores and experimental gases, the adsorption isotherms of CH₄ and N₂ at several temperatures were observed to calculate the adsorption heat and effect of pressure during the single-gas permeation of He, CH₄, and N₂. Furthermore, binary gas separation experiments with He/CH₄ and He/N₂ at ambient temperature and –115 °C were conducted to investigate the degree of adsorption blocking.

2. Experiments

2.1. Sol preparation

BTESE was acquired from Gelest, Inc. as a precursor, without any additional purification. A BTESE-acid sol with sub-nano pores was prepared for membrane fabrication by stirring BTESE, water, and HCl in the ratio of 1:240:10⁻²–10⁻¹ in an ethanol bath at 50 °C for 1 h to promote hydrolysis and condensation reactions. The BTESE concentration was controlled at 5 wt% [34].

A BTESE-swing sol with nanoporous layer for membrane fabrication was prepared by following the steps described in a previous publication [35]. NH₃ and HCl were used as catalysts to adjust the pH of the BTESE-swing sol. First, a mole fraction of BTESE/H₂O/HCl in the ratio of 1:240:10⁻²–10⁻¹ was mixed in an aqueous solution at 50 °C for 1 h. Next, NH₃ was added to the BTESE-acid sol such that the NH₃/HCl molar fraction was 6; this was maintained at 25 °C for 40 min to ensure an alkaline pH to promote condensation. Finally, HCl was added again to the solution such that the molar fraction of HCl/NH₃ was 0.8 to return the solution to an acidic pH, thereby obtaining a stable BTESE-swing sol. The BTESE concentration was controlled at 5 wt%.

2.2. Membrane fabrication

The membranes considered in the experiments were fabricated ac-

cording to published literature [36,37]. Porous α -alumina tubes (average pore size = $1\ \mu\text{m}$; outer diameter = $10\ \text{mm}$; length = $100\ \text{mm}$; Nikkato Co.) were employed as supports for the fabrication of the separation layers. First, α -alumina particles (dimeters of $1\text{--}2\ \mu\text{m}$ and $0.2\ \mu\text{m}$) diluted to 5–10 wt% using the BTESE-swing sol were prepared on the surface of the α -alumina tube to smoothly narrow the pore size and then calcined at $300\ ^\circ\text{C}$ under N_2 . This procedure was repeated several times to obtain the particle layer. Then, the BTESE-swing sol (diluted to approximately 0.5 wt%) was coated onto the particle layer via a hot-coating method at $180\ ^\circ\text{C}$, followed by calcination at $300\ ^\circ\text{C}$ under N_2 to obtain a nanoporous layer. Finally, the BTESE-acid sol that was diluted to 0.25 wt% using ethanol was coated on the nanoporous layer and calcined at $300\ ^\circ\text{C}$ under N_2 ; this process was repeated several times to form a membrane with sub-nano pore size.

2.3. Characterization of BTESE-derived gels

BTESE powders were obtained by slow-drying BTESE-acid sols (BTESE/ $\text{H}_2\text{O}/\text{HCl} = 1/240/10^{-2}$) immersed in containers at $50\ ^\circ\text{C}$ and then calcined at $300\ ^\circ\text{C}$ under N_2 , similar to the conditions used in the membrane fabrication process. N_2 and CH_4 adsorption isotherms of the calcined powders were obtained (BELSOAP MAX, MICROTRAC, Japan) to investigate the adsorption properties of the BTESE powder. The adsorption heats of N_2 and CH_4 were calculated by referring to the observed adsorption isotherms at 200, 240, and $273\ \text{K}$. A mixture of dry ice, a mixture of ice and magnesium chloride, and water cooled via a water circulation system were selected as the coolants for the different temperatures of the adsorption isotherms, that is, 200, 240, and $273\ \text{K}$, respectively.

2.4. Single- and binary gas permeation experiments

All gas permeation experiments were performed under pressurized conditions in the feed side of membranes, as illustrated in Fig. 1. Several gases (He , H_2 , N_2 , CH_4 , CF_4 , and SF_6), with a wide range of kinetic diameters and properties, were selected and fed to the membrane. The quality grade of CH_4 gas was G1. The feed flow rate was regulated using mass flow controllers (Azbil Corporation, Japan). The pressure at the retentate side was regulated using a back pressure valve (Swagelok, USA); the permeate side was maintained at atmospheric pressure. The permeate and retentate sides were connected to the line, in which a soap film meter (HORIBA, Japan) and gas chromatography (GC) system using a thermal conductivity detector (carrier gas: Ar , temperature: $80\ ^\circ\text{C}$, column: MS-5A, GL Science, Japan) were set up. To rapidly exhaust the remaining gas that was fed in a previous experiment, a dry pump (ULVAC, Japan) with a three-way valve was attached to the permeate and retentate sides to change the flow direction.

In the experiments to determine the dependence of gas permeance on kinetic diameter, the pressure in the retentate side was controlled at approximately 130–200 kPa-a using a back pressure valve. Membranes were maintained at $200\ ^\circ\text{C}$ in a mantle heater (purchased from Tokyo National Technical Research Institute, Japan). The feed flow rate was

regulated to maintain the pressure in the retentate side. The flow rate in the permeate side was measured using a soap film meter, and the permeance was calculated using the area of a known membrane surface, flow rate, and pressure difference between retentate and permeate sides. To measure temperature dependence, the permeation temperature was varied from -115 to $200\ ^\circ\text{C}$. Permeation temperatures above the ambient temperature were obtained using a mantle heater. Temperatures below the ambient temperature were achieved by regulating the feed of liquid N_2 via a heat exchanger that was submerged in a vessel filled with freon (AK225G; AGC, Japan), whose freezing point was $-115\ ^\circ\text{C}$. Each low temperature in this experiment was confirmed by monitoring the temperature placed in the vessel until this value was maintained for a couple of minutes. Liquid N_2 was fed from a liquid gas container.

Binary separation experiments were conducted by adjusting the mole fraction of the gas mixture. Two mass flow controllers were used in the binary gas permeation experiments to feed the different gases. The feed flow rate was regulated from 0.4 to $2\ \text{L min}^{-1}$ such that the ratio of the permeate to feed flow rates, which is called stage cut, was below 0.33 to prevent concentration polarization. The pressure at the retentate side was maintained in the range of 400–600 kPa-a using a back pressure valve. The pressure in the permeate side was maintained around atmospheric pressure and no purge gases were used to circulate the permeate gas. The experiments were conducted to achieve He/CH_4 and He/N_2 separation, at varying ratios of the co-existing component with He . The mole fraction and flow rate in the retentate and permeate sides were observed by alternately using the GC system and soap film meter.

3. Results and discussion

3.1. Effect of membrane pore size on single-gas permeation properties

To evaluate the effect of pore structure on the separation mechanism of He , CH_4 , and N_2 , two membranes with different pore sizes were fabricated. The dependence of gas permeance on kinetic diameter at $200\ ^\circ\text{C}$ for BTESE membranes with different pore sizes is shown in Fig. 2 (a) and the dependence of dimensionless gas permeance on the kinetic diameter based on He permeance is shown in Fig. 2 (b). The black-dashed line shown in Fig. 2 (b) denotes the dimensionless permeance of each gas based on He determined according to Knudsen diffusion. Membrane 1 (Memb-1) has small pores that enabled the separation of He , CH_4 , N_2 , and H_2 through molecular sieving. The He permeance was $4.1 \times 10^{-7}\ \text{mol m}^{-2}\ \text{s}^{-1}\ \text{Pa}^{-1}$ and He/CH_4 and He/N_2 selectivity values were 630 and 320, respectively. Note that SF_6 permeance was below the detection limit owing to the presence of small pores. The drastic decrease in permeance of N_2 compared to that of H_2 indicated that this membrane had small pores with sharp pore distributions; this is applicable to He separation conditions over ambient temperatures via enhanced molecular sieving properties.

In contrast, Membrane 2 (Memb-2) has large pores owing to the membrane structure controlled by the nanoporous BTESE layer. Owing to the large pore size, this membrane showed higher He permeance (6.5

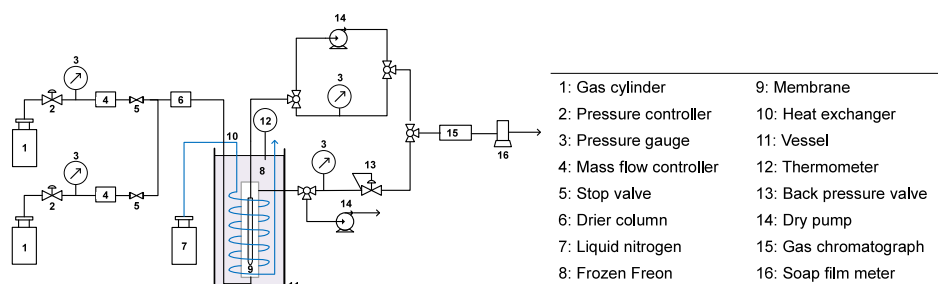


Fig. 1. Schematic diagram of equipment used in the gas separation experiments [33].

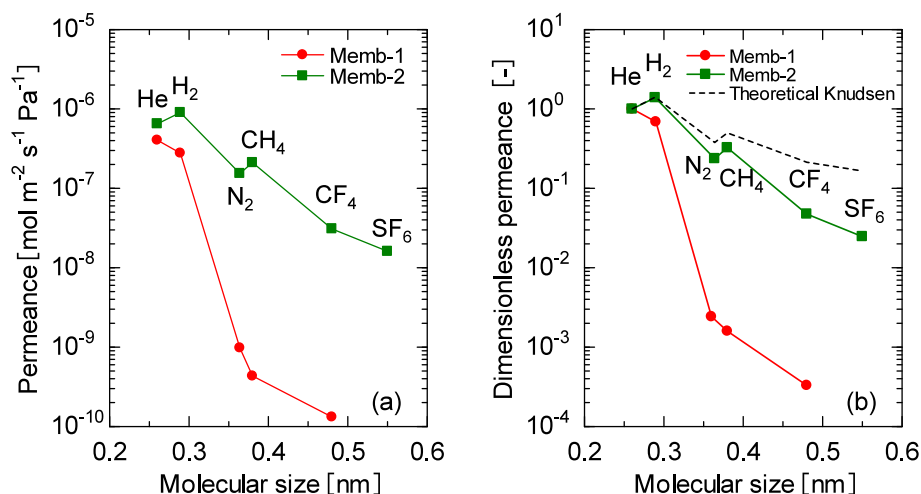


Fig. 2. (a) Dependence of gas permeance on kinetic diameter at 200 °C and (b) dimensionless gas permeance on the kinetic diameter based on He permeance at 200 °C for BTESE membranes with different pore sizes. The black-dashed line in (b) indicates the calculated dimensionless permeance based on Knudsen diffusion for He.

$\times 10^{-7} \text{ mol m}^{-2} \text{s}^{-1} \text{Pa}^{-1}$) than that of Memb-1. However, from Fig. 2 (b), the gas permeance dependence of Memb-2 on the kinetic diameter was approximately similar to the theoretical Knudsen ratio (He/CH₄ selectivity: 2.0 and He/N₂ selectivity: 2.6), except for large molecules such as CF₄ and SF₆. Consequently, compared to Memb-1, this result reinforces that Memb-2 has large pores, which are too large to separate even CH₄ and N₂ via molecular sieving. To quantitatively evaluate the pore size, a modified gas translation (m-GT) model [38], which was derived from the gas translation model [39,40], was used to evaluate the average pore sizes of these membranes, as described in detail in Text S1 and shown in Fig. S1. According to the analysis in Fig. S1, the pore sizes of Memb-1 and Memb-2 are 0.41 and 0.90 nm, respectively.

The temperature dependence of single-gas permeance at temperatures ranging from -115 to 200 °C for BTESE membranes with different pore sizes is shown in Fig. 3. For Memb-1, He and H₂ showed a strong dependence on temperature owing to activated diffusion at temperatures ranging from -50 to 200 °C, resulting in a significant decrease in permeance with decreasing temperature. However, the H₂ and He

permeances eventually deviated from the extrapolation line of their temperature dependences below -50 and around -100 °C, respectively. In contrast, CH₄ and N₂ permeance tended to primarily exhibit Knudsen diffusion at ambient temperature, although a marginal tendency of activated diffusion was observed at approximately 200 °C. At low temperatures, the influences of both Knudsen and surface diffusions were observed. The high He/CH₄ and He/N₂ selectivity values, which were 630 and 320 at 200 °C, respectively, reduced to 4 at -115 °C. This was ascribed to the drastic reduction in the activated-diffusion-induced He permeance, whereas the increase in CH₄ and N₂ permeances with the decrease in permeation temperatures was induced by Knudsen diffusion. Consequently, He selectivity deteriorated at -115 °C. Note that the reduction in temperature caused a diversion of He and H₂ permeance from the extrapolation line owing to the limitations in He and H₂ diffusions through small pores at low temperatures. Hence, this permeation condition did not provide the necessary energy for He and H₂ to diffuse into the small pores and forced He and H₂ to permeate through grain boundaries via Knudsen diffusion at extremely low temperatures.

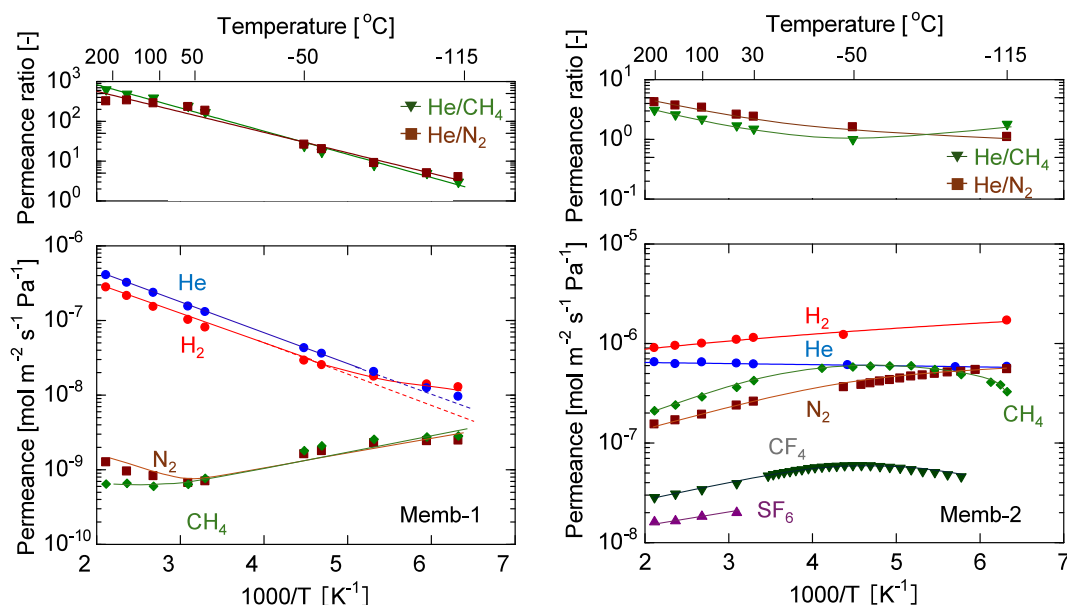


Fig. 3. Temperature dependence of single-gas permeance (He, H₂, N₂, CH₄, CF₄, SF₆) and permeance ratio at temperatures ranging from -115 to 200 °C for BTESE membranes with different pore sizes (Memb-1 and Memb-2).

Consequently, the He and H₂ permeances drastically decreased from the order of 10^{−7} to 10^{−9} and approached the Knudsen ratio relative to CH₄ and N₂. This result indicated that small pores, which show superior molecular sieving effects to achieve high separation performance at ambient or elevated temperatures, could not demonstrate practical He separation performance at low temperatures.

Conversely, for Memb-2, which had larger pores than those of Memb-1, H₂ permeance exhibited Knudsen diffusion over the entire range of permeation temperatures. The degree of activated diffusion of He was alleviated because of the large pore sizes compared to those of Memb-1. CH₄ and N₂ permeances displayed evident Knudsen diffusion from 30 to 200 °C. It should be noted that, considering the operating pressure in the experiments, the boiling point of SF₆ was closer to 0 °C compared with those of the other gases, including CF₄ (Tables S1 and S2). Owing to safety issues in the experiment and to adhere to the law, in the present study, the temperature dependence of gases was evaluated in the range where condensation is not likely to occur thermodynamically. Therefore, the temperature dependence of SF₆ below ambient temperature, which is approximately 30 °C, was not evaluated. Intriguingly, CH₄ and CF₄ permeances reached a local maximum value around −50 °C, which is around the critical temperatures of these gases (191 K for CH₄ and 228 K for CF₄), and decreased with further reductions in temperature. N₂ permeance deviated from the extrapolation of its Knudsen line. According to previous studies [41,42], surface diffusion dominates the separation property during the permeation of adsorptive molecules such as CH₄ and N₂ at extremely low temperatures. This experimental result reinforced that CH₄ and N₂, which are adsorptive molecules, permeated through membrane pores via surface diffusion at extremely low temperatures, leading to self-permeate inhibition.

3.2. Effect of operation pressure on adsorption and gas permeation

The adsorption heats of CH₄ and N₂ for various inorganic materials [43–45] including BTESE powders are summarized in Table 1. The adsorption heats of CH₄ and N₂ for BTESE were estimated using the Clausius–Clapeyron equation considering the adsorption heat isotherm at different temperatures (200, 240, and 273 K), as shown in Fig. S2. According to the adsorption isotherms at different temperatures, a sharper increase in the amount of adsorption of CH₄ and N₂ was observed with the reduction in adsorption temperature. A drastic increase in adsorption amount at low pressure has often been observed in microporous materials. The obtained adsorption heat of CH₄ for BTESE was larger than that of N₂. Note that the adsorption heat of He is quite small owing to the significantly low boiling temperature of 4 K [29]. Aside from adsorption heat, the adsorption mechanism depending on the membrane materials is also an influential parameter. Generally, zeolite-like materials tend to show high adsorbed loading compared to that of amorphous silica-like materials.

Fig. 4 shows the feed pressure dependence of single-gas permeance (He, CH₄, and N₂) and dimensionless permeance at 30 and −115 °C for the membranes with different pore sizes (Memb-1 and Memb-2). To obtain the dimensionless permeance for each gas in Fig. 4, the observed gas permeances at pressures ranging from 200 to 600 kPa were divided (normalized) by the gas permeance measured at 200 kPa, the lowest feed pressure used in the present study. If the decrease in the dimensionless

permeance ratio is greater, a larger adsorption-blocking effect is expected during binary separation. At −115 °C, the permeance of CH₄ and N₂ through Memb-2 reduced by 30–40% with the increase in feed pressure. Conversely, at −115 °C, feed pressure did not exhibit any influence on permeance through Memb-1. At 30 °C, none of the experimental gases exhibited pressure dependence for both Memb-1 and Memb-2. The experimental gas permeance was not dependent on feed pressure at −115 °C for Memb-1 because the pore size of Memb-1 was too small to achieve surface diffusion of the gases. Thus, these experimental gases, including CH₄, could not form an adsorption layer. In contrast, Memb-2 has large pores that facilitate the adsorption of CH₄ and N₂. Note that the permeance of He was unaffected by feed pressure, irrespective of the permeation temperature and membrane pore size, even though CH₄ and N₂ permeances were affected by pressure variations. This result can be ascribed to Henry's law of adsorption and the low standard boiling temperature of He of 4 K.

Furthermore, for Memb-2 at −115 °C, the degree of dependence on feed pressure for CH₄ was larger than that for N₂. This result is attributed to the difference in their adsorptive properties, as confirmed in Table 1. Although self-inhibition during single-gas permeation through the network pores occurs in CH₄ and N₂, the greater adsorptive property for CH₄ compared to that of N₂ accelerates the extensive growth of a high-density phase in the network pores. Moreover, as the characteristics of the high-density phase are similar to a liquid, the difference in critical temperature also significantly contributes to the difference in the feed pressure dependence between CH₄ and N₂. That is, CH₄, whose critical temperature was included in the observed temperature region, was influenced by the formation of a high-density layer when compared to N₂, whose critical temperature was lower than the lowest measurement temperature (−115 °C).

3.3. Binary separation properties of natural gas components for BTESE membranes

Binary separation experiments were conducted for Memb-1 and Memb-2. To confirm the stability, accuracy, and reliability of the method used for obtaining permeation data of binary separation, particularly at low temperatures, time courses of He and CH₄ permeance at −115 °C in single and binary separations and He and N₂ permeance in single and binary separations for Memb-2 are shown in Fig. 5. Note that single-gas permeance was measured in the first stage for each separation system and CH₄ and N₂ was introduced by progressively changing the mole fraction of the component co-existing with He. After binary separation, single-gas permeance was measured for membrane stability and physical adsorption of CH₄ and N₂. Note that the feed flow rate was regulated from 0.9 to 1.6 L min^{−1} to secure a stage cut of 0.21–0.33. Figs. 3 and 4 clarify that, in single-gas permeation, Memb-2 shows a He permselective property. However, once CH₄ or N₂ was introduced in the binary system, He permeance dropped significantly in contrast to the increase in the permeance for CH₄ and N₂, which surpassed He permeance. The selectivity of CH₄/He and N₂/He in each binary system increased gradually up to 10 and 5.5, respectively, with an increase in mole fraction of CH₄ or N₂ in the feed. Note that the permeance of He, CH₄, and N₂ in single-gas permeation after binary separation recovered to that achieved in the pristine condition. Moreover, the permeance of each gas at any co-existing component fraction was stable, confirming the membrane stability and accuracy of each permeation data.

The effects of different He mole fractions in the feed side on He, CH₄, and N₂ permeance and He selectivity for Memb-1 at 50 and −115 °C in the binary separation for (a) He/CH₄ and (b) He/N₂ are shown Fig. 6. The data of single-gas permeation and binary separation for He/CH₄ and He/N₂ at 50 °C indicate that He, CH₄, and N₂ permeance were unaffected by the He mole fraction. In the binary separation at 50 °C, Memb-1 shows high He permeance of 1.5 × 10^{−7} mol m^{−2} s^{−1} Pa^{−1} with He/CH₄ and He/N₂ selectivity values of approximately 260. However, in contrast to the result of single-gas permeation, as opposed to the

Table 1
Comparison of adsorption heats of CH₄ and N₂.

Membrane	Adsorption heat [kJ mol ^{−1}]	
	CH ₄	N ₂
BTESE (this study)	19.1	11.5
Zeolite (5A) [43]	14.8	-
MFI [44]	18	-
Carbon molecular sieve [45]	18.7	16.9
Silicalite-1 [29]	22.6	13.8

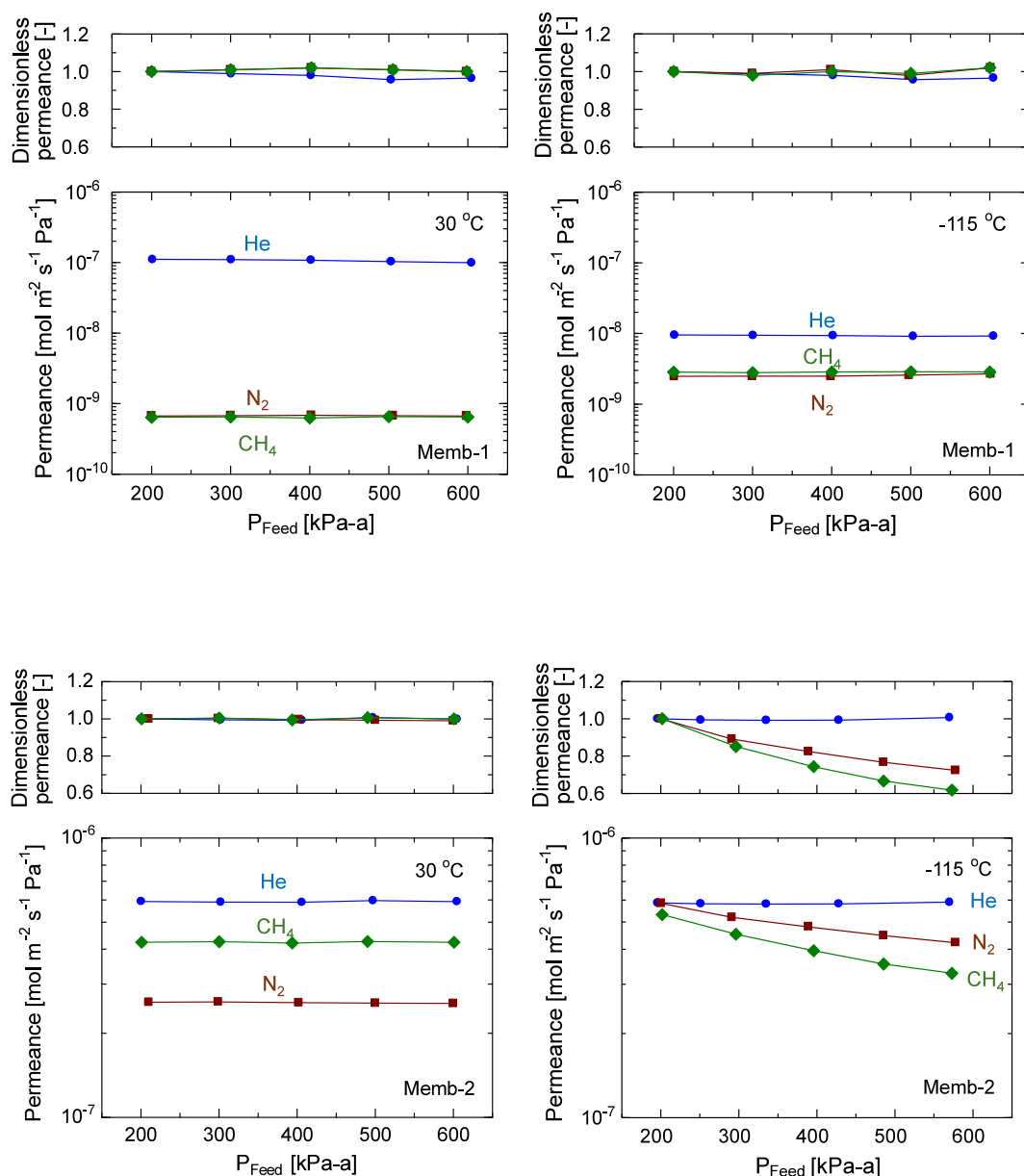


Fig. 4. Feed pressure dependence of single-gas permeance (He, CH₄ and N₂) at 30 and -115 °C for BTESE membranes with different pore sizes (Memb-1 and Memb-2).

constant permeance of CH₄ and N₂ at -115 °C for the different He mole fractions in the feed side, He permeance in the binary system was smaller than that in single-gas permeation. This was due to the deterioration of He diffusivity by the presence of a co-existing gas with lower diffusivity. According to relevant studies [46,47], the effect of a high-adsorptive and large-molecular co-existing component with a non-adsorptive one was demonstrated for microporous silica membranes and metal-organic frameworks. In summary, He permeance and selectivity decreased significantly to the order of 10⁻⁹ and approximately 1, respectively, indicating that a membrane with small pores, such as Memb-1, is inappropriate for He separation when in the presence of adsorptive co-existing components such as CH₄ and N₂ at extremely low temperatures.

The effects of the different He mole fractions in the feed side on He, CH₄, and N₂ permeance and He selectivity using Memb-2 at 50 and -115 °C in the binary separation of (a) He/CH₄ and (b) He/N₂ are shown in Fig. 7. As observed in the binary separation of He/CH₄ and He/N₂ at 50 °C, He, CH₄, and N₂ permeances were unaffected by the He

mole fraction in the feed side. Thus, the preferential permeation component, which was He, was identical in both cases. However, as opposed to the binary separation of He/CH₄ and He/N₂ at -115 °C using Memb-1, the preferential permeation component changed from He to CH₄ or N₂ in the binary separation (CH₄/He: 10, N₂/He: 5.5) of He/CH₄ and He/N₂ using Memb-2 at -115 °C. This was due to the occurrence of the surface-diffusion-induced high-density phases of CH₄ or N₂. He permeance decreased with the increase in CH₄ or N₂ mole fraction. This indicates that the high-density phase of CH₄ or N₂ under high mole fraction in the feed inhibited He permeation.

The relationship between the He mole fractions in the feed and permeate sides for Memb-1 at 50 and -115 °C in binary He/CH₄ and He/N₂ separation are shown in Fig. 8. The theory of predicted separation curve and separation limit was described in a previous study [48] and is elaborately discussed in Supplementary Information. Note that the observed experimental data of separation performance for Memb-1 at 50 °C is not presented minutely unlike that at -115 °C. This is because of the restrictions in the experimental conditions and separation

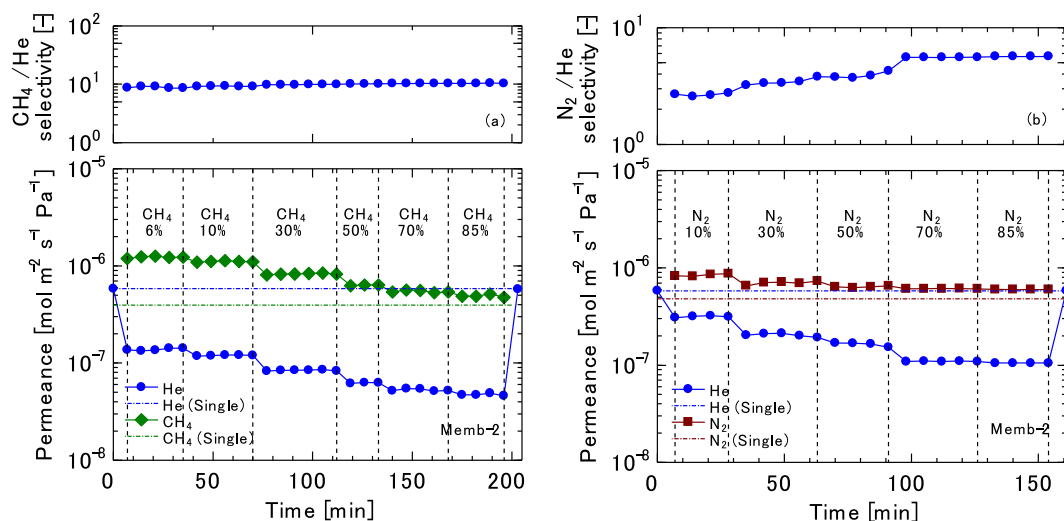


Fig. 5. Time courses at $-115\text{ }^{\circ}\text{C}$ of (a) He and CH₄ permeance in binary separation and (b) He and N₂ permeance in binary separation for Memb-2. The feed pressure was maintaining at 400 kPa-a; the permeate side was at atmospheric pressure.

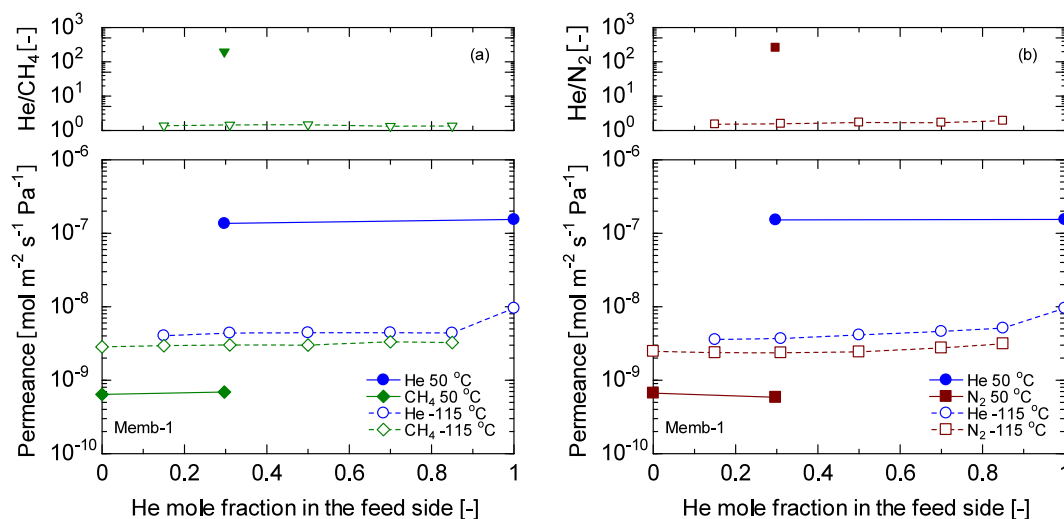


Fig. 6. Effect of different He mole fractions in the feed side on He, CH₄, and N₂ permeance and He selectivity for Memb-1 at 50 and $-115\text{ }^{\circ}\text{C}$ in the binary separation for (a) He/CH₄ and (b) He/N₂. The feed pressure was maintained at 400 and 600 kPa-a at 50 and $-115\text{ }^{\circ}\text{C}$, respectively. The permeate side was maintained at atmospheric pressure. The feed flow rate was regulated to approximately 0.4 L min^{-1} .

performance of Memb-1. The small structure of Memb-1 enhanced the molecular sieving effect, as this resulted in a small flow rate in the permeate side, the predicted separation curves of He/CH₄ and He/N₂ were relatively close to the separation limit and the upper horizontal axis of around <20% and >40 % He mole fraction in the feed side. Consequently, only one data for He/CH₄ and He/N₂ at 50 °C was observed around 30 % He mole fraction in the feed side to confirm whether the He separation property of Memb-1 at 50 °C was affected by the presence of CH₄ or N₂ in the feed. Regarding the separation performance of Memb-1 for He/CH₄ and He/N₂ at 50 °C, the observed data were nearly on the predicted separation curve. Given the results of feed pressure dependence of single-gas permeance for Memb-1 at 50 °C, we can consider that He permeation was not influenced by the permeation of CH₄ or N₂ in the binary separation experiments. This result reinforces the fact that the dominant permeation property at 50 °C is the molecular sieving effect and the effects related to the interaction of the pore surfaces of membranes and gases are not observed.

However, interestingly, unlike the experimental results at 50 °C, the separation performance of He/CH₄ and He/N₂ at $-115\text{ }^{\circ}\text{C}$ varied compared to that at 50 °C. The observed results of He/CH₄ and He/N₂

for Memb-1 at $-115\text{ }^{\circ}\text{C}$ appeared to deviate from the predicted separation curve by 10–30% such that the He mole fraction in the permeate side was lower than that of the predicted separation curve based on the He mole fraction in the feed side. Notably, for both He/CH₄ and He/N₂, the preferential permeation component was He, irrespective of the permeation mode (single, binary). This result indicates that reduction in the He mole fraction in the permeate side can be ascribed to the reduction in He permeation due to the presence of co-existing components in the gas phase. Note that the reduction in He mole fraction in the permeate side was due to the mutual interaction of permeating gas molecules and not by the high-density phase of co-existing components such as CH₄ and N₂ with He [46,47].

The relationship between the He mole fractions in the feed and permeate sides in the binary separation of He/CH₄ and He/N₂ using Memb-2 at 50 and $-115\text{ }^{\circ}\text{C}$ are shown in Fig. 9. At 50 °C, the observed data for He/CH₄ and He/N₂ over the entire range of He mole fractions were on the predicted separation curve, indicating that the separation mechanism was dominated by molecular sieving, irrespective of the membrane pore size. Conversely, the separation performance of He/CH₄ and He/N₂ at $-115\text{ }^{\circ}\text{C}$ showed a remarkable difference compared to that

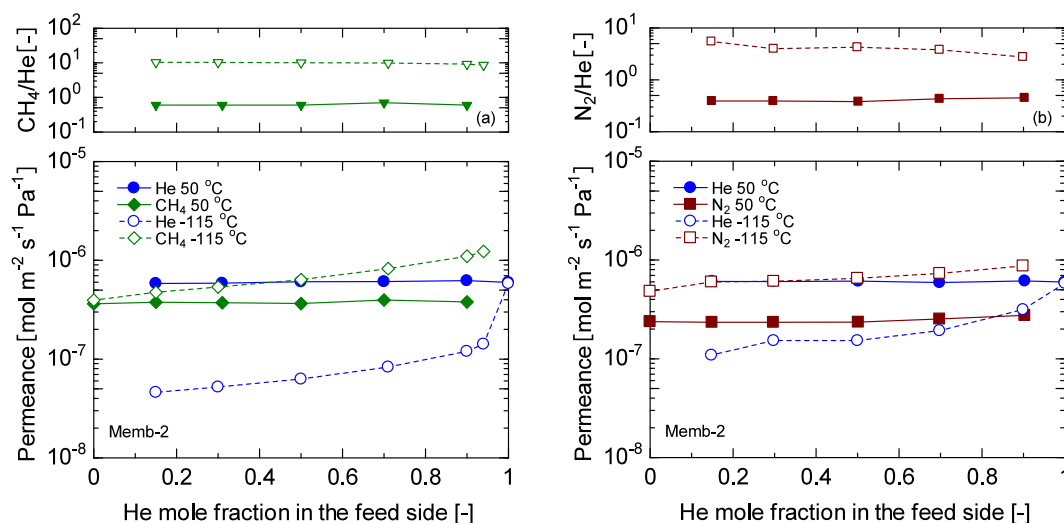


Fig. 7. Effects of different He mole fractions in the feed side on He, CH₄, and N₂ permeance and He selectivity using Memb-2 at 50 and –115 °C in the binary separation of (a) CH₄/He and (b) N₂/He. The feed pressure was maintained at 400 kPa-a at both 50 and –115 °C. The permeate side was maintained at atmospheric pressure. The feed flow rate was regulated to approximately 0.8 to 1.6 L min^{–1}.

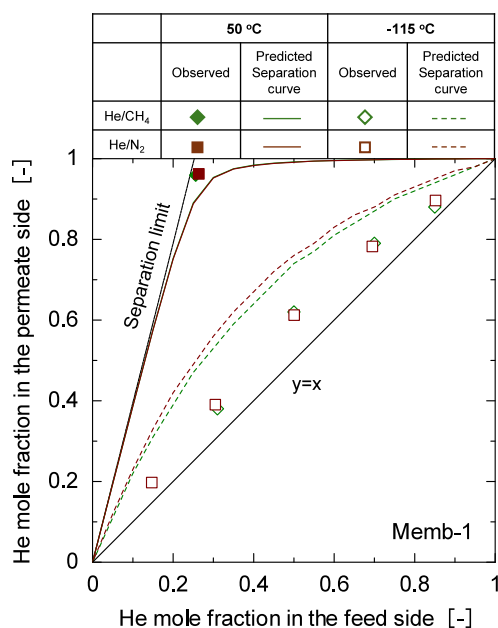


Fig. 8. Relationship between the He mole fractions in the feed and permeate sides for Memb-1 at 50 and –115 °C in binary He/CH₄ and He/N₂ separation. The feed pressure was maintained at 400 and 600 kPa-a at 50 and –115 °C, respectively. The permeate side was maintained at atmospheric pressure. The feed flow rate was regulated to approximately 0.4 L min^{–1}.

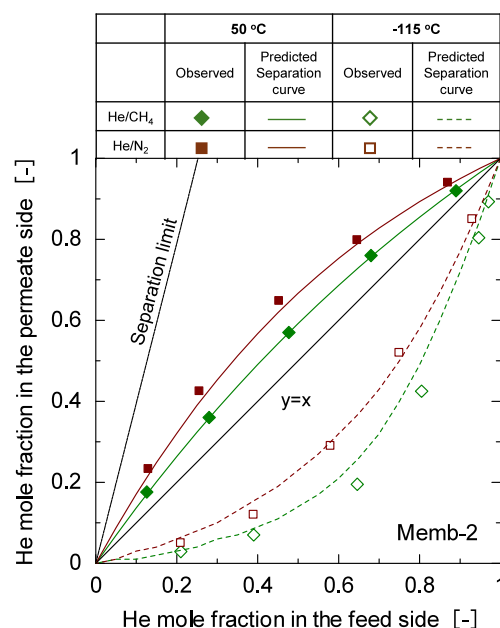


Fig. 9. Relationship between the He mole fractions in the feed and permeate sides in the binary separation of He/CH₄ and He/N₂ using Memb-2 at 50 and –115 °C. The feed pressure was maintained at 400 kPa-a at 50 and –115 °C and the permeate side was maintained at atmospheric pressure. The feed flow rate was regulated from 0.8 to 1.6 L min^{–1}.

for Memb-1 as the He mole fraction in the permeate side was smaller than that in the feed side. Furthermore, the He mole fraction in the permeate side for He/CH₄ separation was smaller than that for He/N₂ separation. This observation indicates that the degree of adsorptive blocking caused by high-density adsorptive molecules in the membrane for CH₄ was more influential than that for N₂, as is also evident by the difference in adsorption heat between CH₄ and N₂. This trend is consistent with the results of feed pressure dependence of single-gas permeation (He, CH₄, and N₂) for Memb-2 at –115 °C. Note that the difference of critical temperature and the kinetic diameter of between CH₄ and N₂ must be considered about the degree of adsorptive blocking.

To confirm the He separation property of Memb-2 in the binary separation of He/CH₄ and He/N₂ at –115 °C, for which feed pressure

dependence of single-gas permeation was observed, the CH₄ and N₂ permeances in both single-gas permeation and binary separation experiments as a function of CH₄ and N₂ partial pressures for Memb-2 at –115 °C are simultaneously illustrated in Fig. 10. The CH₄ and N₂ permeances observed in the binary separation were almost overlapped with those observed in single-gas permeation. These results elucidates that the He permeation property is hindered by the presence of adsorbed molecules, but the presence of He do not affect the permeation of adsorptive molecules of N₂ and CH₄.

The conceptual schematic images of the separation properties of the two membranes with different pores at ambient and extremely low temperatures are illustrated in Fig. 11 to summarize the series of experimental results obtained in this study. In terms of the permeation

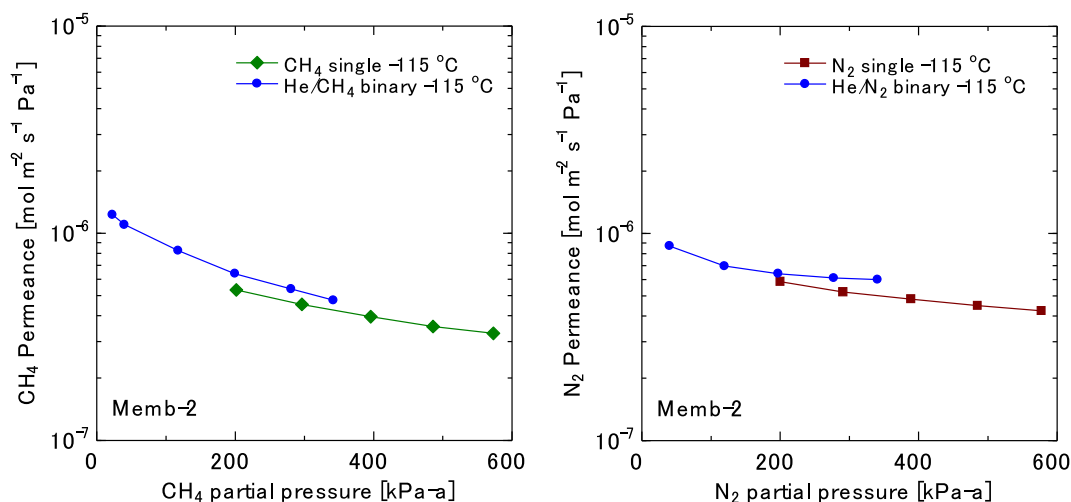


Fig. 10. CH_4 and N_2 permeances in single-gas permeation and binary separation (He/CH_4 , He/N_2) as a function of CH_4 and N_2 partial pressures for Memb-2 at -115°C .

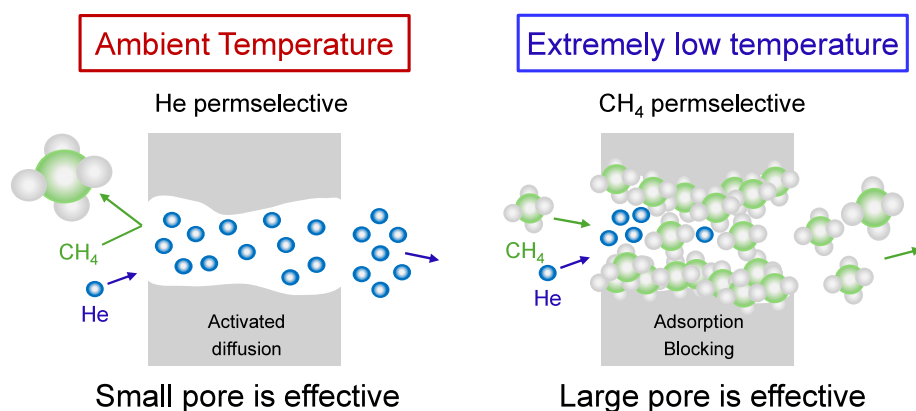


Fig. 11. Conceptual schematic images of the separation properties of two membranes with different pore sizes at ambient and extremely low temperatures.

property of BTESE membranes with small pores at ambient temperatures, high He separation performance was observed owing to the activated diffusion derived from the molecular sieving effect with small pores. However, as temperature decreases, the separation mechanism

was affected by grain boundaries in a membrane that is dominated by Knudsen diffusion. Then, as the temperature decreased to an extremely low temperature, the interference of adsorptive molecules including CH_4 or N_2 to superior He diffusion leads to lower separation performance

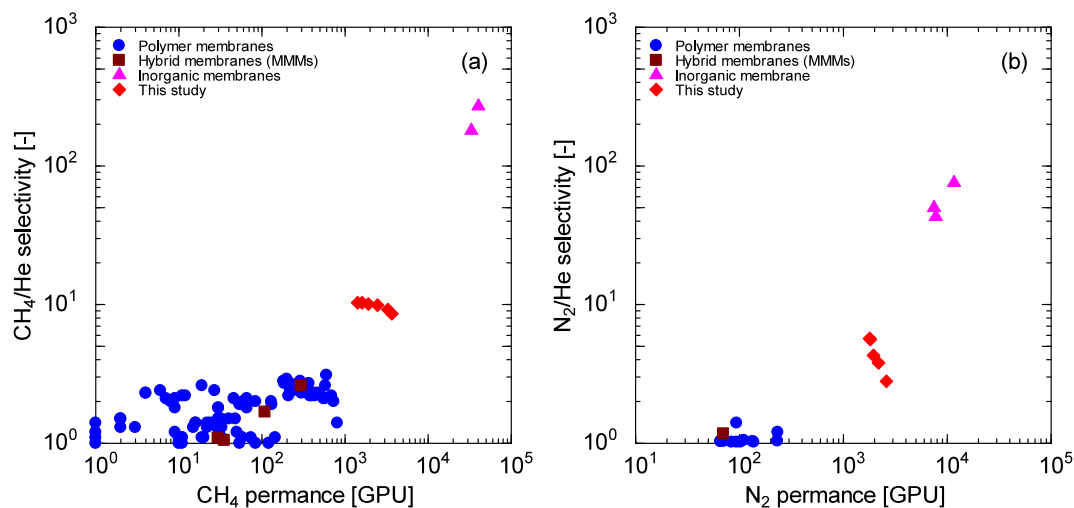


Fig. 12. Trade-off curves of (a) CH_4/He and (b) N_2/He separation using conventional membranes, including polymer, inorganic, and hybrid (MMMs) membranes, and BTESE membranes.

than that expected through Knudsen diffusion. Note that a high-density phase is not formed in small pores at extremely low temperatures as observed in the binary separation of He/CH₄ and He/N₂ at −115 °C.

The Knudsen diffusion is the dominant mechanism in BTESE membranes with large pores at ambient temperatures; consequently, the He permselective property is diminished. However, the high-density phase of CH₄ or N₂, which plays a role in adsorption blocking, is formed in a large structure at extremely low temperatures. Consequently, the preferential permeation component changes from He to CH₄ or N₂.

3.4. Comparison between BTESE and conventional membranes with preferential permeation of N₂ and CH₄ against He

To evaluate He separation performance of various membrane materials, trade-off curves of (a) He/CH₄ and (b) He/N₂ separation using conventional membranes, including polymer, inorganic, and hybrid (MMM) membranes, and BTESE membranes (refer to Tables S3 and S4 for detailed membrane data), are shown in Fig. 12. The permeabilities of CH₄ and N₂ in the literature [9] were measured in Barrer. These values were recalculated by dividing the original permeability by the average reported membrane thickness to obtain the measurements in GPU. For polymer and hybrid membranes, CH₄/He selectivity was 3 and CH₄ permeance reached approximately 1000 GPU. N₂/He selectivity was 2 and N₂ permeance was a maximum of 200 GPU. Even though conventional polymer membranes showed CH₄ or N₂ permselective property via solution-diffusion mechanism, according to the literature, the number of studies to show high CH₄ or N₂ permselective was limited. In contrast, BTESE membranes in this study exhibited moderate separation performance with CH₄/He selectivity of 10 and N₂/He selectivity of 6 at −115 °C, as well as high CH₄ and N₂ permeance of $1.2 \times 10^{-6} \text{ mol m}^{-2} \text{ s}^{-1} \text{ Pa}^{-1}$ (3600 GPU) and $8.7 \times 10^{-7} \text{ mol m}^{-2} \text{ s}^{-1} \text{ Pa}^{-1}$ (2600 GPU), respectively. The separation performance for MFI-type zeolite membranes was exceptional. CH₄/He and N₂/He selectivity were 270 and 70, respectively. CH₄ and N₂ permeance was $1.4 \times 10^{-5} \text{ mol m}^{-2} \text{ s}^{-1} \text{ Pa}^{-1}$ (40300 GPU) and $3.9 \times 10^{-6} \text{ mol m}^{-2} \text{ s}^{-1} \text{ Pa}^{-1}$ (11640 GPU) [30,31]. As opposed to that conventional inorganic membranes show He permselective property because of the smaller molecular size of He than those of N₂ and CH₄, the reason for CH₄ or N₂ permselective property can be ascribed to the blocking effect of CH₄ or N₂ adsorption.

To clarify the effect of permeation temperature on CH₄/He or N₂/He selectivity, temperature dependencies of (a) CH₄/He and (b) N₂/He selectivity when using conventional membranes, including polymer, inorganic, and hybrid (MMMs) membranes, and BTESE membranes are shown in Fig. 13. Compared to the number of studies on He

permselective membranes, the number of studies on CH₄ or N₂ permselective membranes is significantly limited. This is because achieving CH₄ or N₂ permselective permeation for separating He is challenging, particularly at extremely low temperatures. The CH₄/He selectivity for polymer and hybrid membranes were plotted around 3 on the horizontal axis, which implies that the temperature was around 300 K, whereas that for inorganic membranes including BTESE membranes in this study ranged from 5.5 to 8 of the horizontal axis, which implies that the temperature ranged from 124 to 180 K. This observation demonstrates that CH₄ and N₂ selectivity are improved with the decrease in permeation temperature. This is ascribed to the enhancement of surface diffusion at low temperatures.

Even though the CH₄ or N₂ permselective properties can be enhanced by modulating the operating conditions (e.g., the feed pressure, temperature, and concentration), these new findings indicate that a well-ordered rigid structure, such as a crystalline zeolite, is appropriate for enhancing the adsorption-blocking effect. The reason for this is not clear at the moment. However, one possibility is that the permeation path may be restricted in the case of a membrane with a uniform pore-size distribution, such that the adsorbed molecules effectively block the permeation of non-adsorptive molecules. Thus, in terms of the structural design of amorphous materials, further studies should focus on the design of uniform structures with enhanced adsorption properties via the microstructural engineering of amorphous silica, such as through ion doping, metal oxide incorporation to form composite ceramics, and fabricating carbon-containing ceramic composites [49].

4. Conclusions

He separation performance of mixtures derived from natural gas, which include He, CH₄, and N₂, at extremely low temperatures was investigated using BTESE bridged-type organosilica membranes with different pore sizes. The membrane pore size was successfully controlled through the sol preparation conditions (pH swing, acid/water molar ratio). A BTESE membrane with small pores showed high He selectivity (i.e., He/CH₄: 630, He/N₂: 320 at 200 °C) from 30 to 200 °C owing to enhanced molecular sieving. However, the He permeance through this membrane decreased as the permeation temperature decreased and He selectivity dropped to the Knudsen ratio at −115 °C. Conversely, a BTESE membrane with relatively large pores showed low He selectivity from 30 to 200 °C; however, this membrane showed CH₄ and N₂ permselective (CH₄/He: 10, N₂/He: 6 at −115 °C) properties, as well as high CH₄ and N₂ permeance of $1.2 \times 10^{-6} \text{ mol m}^{-2} \text{ s}^{-1} \text{ Pa}^{-1}$ (3600 GPU) and $8.7 \times 10^{-7} \text{ mol m}^{-2} \text{ s}^{-1} \text{ Pa}^{-1}$ (2600 GPU), at −115 °C, respectively.

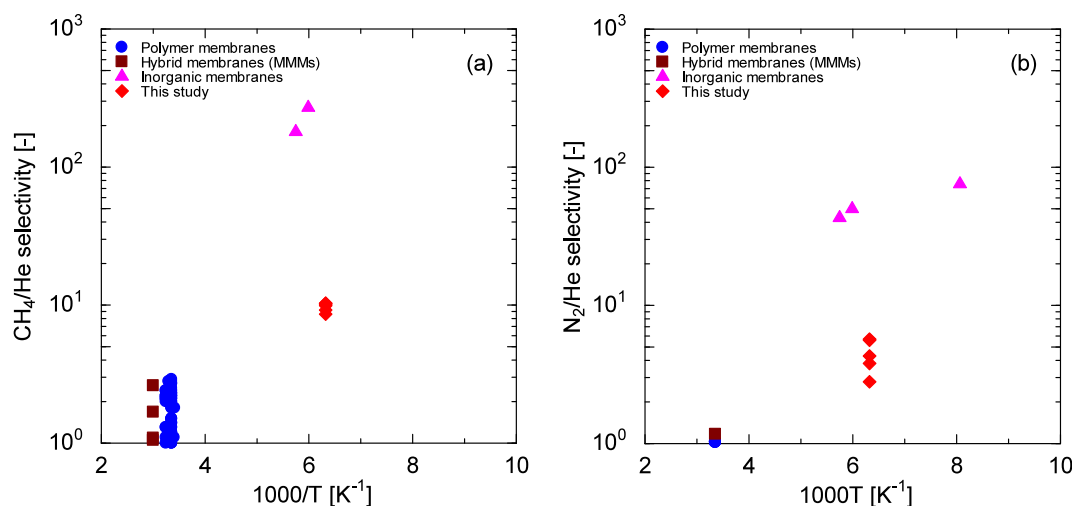


Fig. 13. Temperature dependencies of (a) CH₄/He and (b) N₂/He selectivity when using conventional membranes, including polymer, inorganic, and hybrid (MMMs) membranes, and BTESE membranes.

This was caused by surface diffusion of CH₄ and N₂ at extremely low temperatures, thereby effectively inhibiting the permeation of He, which has a smaller molecular size than those of CH₄ and N₂, through adsorption blocking. This effect was strongly confirmed for the membrane with large pores. These results shed light on the separation properties of an adsorptive/non-adsorptive system at extremely low temperatures and recommend that the use of a membrane with large pore sizes is effective, as opposed to small pores, which is the appropriate design at ambient temperature, to exploit the molecular sieving.

CRedit authorship contribution statement

K. Ishizaki: Writing – original draft, Resources, Methodology, Data curation. **K. Sato:** Investigation. **N. Moriyama:** Writing – review & editing. **H. Nagasawa:** Writing – review & editing. **T. Tsuru:** Writing – review & editing. **M. Kanezashi:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

The authors declare that they have not used generative AI and AI-assisted technologies in the writing process.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. We declare that no known conflict of interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.memsci.2026.125299>.

Data availability

Data will be made available on request.

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